Pegi Teknik

Tarih:........../........../..........

27.11.2019 (Vize Sonrası 11. hafta)

Ters Z dönüşümü

1- Residü Metodu

2- Kuvvet Serileri → payın derecesi paydadan büyük yada eşit ise uygulanır.

3- Kesirlere Ayırma → klasik matematik problemi

4- Fark Denklemleri → $x(n-1)$ ise $X(z).z^{-1}$

Residü Metodu

$$A = \frac{(z-p_1) \cdot Az}{(z-p_1) \cdots} \Bigg|_{z=p_1} =$$

Örnek: $X(z) = \frac{1}{1-az^{-1}}$ $|z|>|a|$ $x(n)=?$

$1-az^{-1} = 0$

$az^{-1} = 1$

$\boxed{z=a}$ → kutup.

$x(n) = ($

Bu örnek iptal oldu.

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örnek: $X(z) = \dfrac{1}{z^2 \cdot (z-a)} = \dfrac{A}{z^2} + \dfrac{B}{z-a}$

$\left.\begin{array}{l} z=0 \\ z=a \end{array}\right\}$ kutuplar

$A = z^2 \cdot \dfrac{1}{z^2 \cdot (z-a)} \Bigg|_{z=a} = -\dfrac{1}{a}$

$B = \dfrac{(z-a) \cdot 1}{z^2 (z-a)} \Bigg|_{z=a} = \dfrac{1}{a^2}$

$\dfrac{-1/a}{z^2} + \dfrac{1/a^2}{z-a} = \dfrac{1}{a} z^{-2} + \dfrac{1}{a^2} \cdot \dfrac{z^{-1}}{1-a z^{-1}} = \dfrac{1}{a} \delta(n-2) + \dfrac{1}{a^2} \cdot a^{n-1} \cdot U(n-1)$

örnek: $X(z) = \dfrac{z}{\underbrace{2z^2 - 3z + 1}}$ $\quad x(n) = ?$

kökler

$\Delta = 9 - 8 = 1$

$z_1 = \dfrac{3 \mp 1}{4} = 1$

$z_2 = \dfrac{1}{2}$

$\dfrac{z}{2z^2 - 3z + 1} = \dfrac{z}{2 \cdot (z-1)(z-\frac{1}{2})} = \dfrac{1}{2} \cdot \dfrac{z}{(z-1)(z-\frac{1}{2})}$

$= \dfrac{1}{2} \cdot \dfrac{z^{-1}}{(1-z^{-1})(1-\frac{1}{2}z^{-1})} = \dfrac{A}{1-z^{-1}} - \dfrac{B}{1-\frac{1}{2}z^{-1}}$

$A = \dfrac{\cancel{(1-z^{-1})} \cdot \frac{1}{2} \cdot z^{-1}}{\cancel{(1-z^{-1})}(1-\frac{1}{2}z^{-1})} \Bigg|_{z^{-1}=1} = \dfrac{\frac{1}{2}}{\frac{1}{2}} = 1$

$B = \dfrac{\cancel{(1-\frac{1}{2}z^{-1})} \cdot \frac{1}{2} z^{-1}}{(1-z^{-1})\cancel{(1-\frac{1}{2} z^{-1})}} \Bigg|_{z^{-1}=2} = -1$

$= \dfrac{1}{1-z^{-1}} - \dfrac{1}{1-\frac{1}{2} \cdot z^{-1}}$

$= U(n) = \left(\dfrac{1}{2}\right)^n U(n)$

Tarih:........../........../..........

Örnek : $X(z) = \dfrac{\frac{3}{4}}{(1-\frac{1}{2}z)(1-\frac{1}{2}z^{-1})}$ $\quad \left|\frac{1}{2}\right| < |z| < |2|$

$$= \frac{3/4}{(z^{-1}-\frac{1}{2})(1-\frac{1}{2}z^{-1})} = \frac{3/2 \cdot z^{-1}}{(1-2z^{-1})(1-\frac{1}{2}z^{-1})}$$ = hem pay hem paydan 2 ile çarpıldı

$$A = -\cancel{(1-2z^{-1})} \cdot \frac{3/2\ z^{-1}}{\cancel{-}(1-\frac{1}{2}z^{-1})}\Bigg|_{z^{-1}=\frac{1}{2}} = -1$$

$$B = -\frac{3/2\ z^{-1}}{1-2^{z-1}}\Bigg|_{z^{-1}=2} = 1$$

$$-\frac{1}{1-2z^{-1}} + \frac{1}{1-\frac{1}{2}z^{-1}}$$

$$= 2^n u(-n-1) + \left(\frac{1}{2}\right)^n \cdot u(n)$$

Örnek!

$$X(z)=\frac{1}{(1-z^{-1})^2\cdot(1+z^{-1})}=\frac{A}{(1-z^{-1})^2}+\frac{B}{1-z^{-1}}+\frac{C}{1+z^{-1}}$$

$$A=\cancel{(1-z^{-1})^2}\cdot\frac{1}{\cancel{(1-z^{-1})^2}\cdot(1+z^{-1})}\Bigg|_{z^{-1}=1}=\frac{1}{2}$$

$$B=\frac{d}{dz}A(z)=\frac{d}{dz}\left(\frac{1}{1+z^{-1}}\right)\Bigg|_{z^{-1}=1}=\frac{0-(0-z^{-2})}{(1+z^{-1})^2}=\frac{z^{-2}}{(1+z^{-1})^2}\Bigg|_{z^{-1}=1}=\frac{1}{4}$$

$$C=\cancel{(1+z^{-1})}\cdot\frac{1}{(1-z^{-1})^2\cancel{(1+z^{-1})}}\Bigg|_{z^{-1}=-1}=\frac{1}{4}$$

$$=\underbrace{\frac{1/2}{(1-z^{-1})^2}}+\underbrace{\frac{1/4}{1-z^{-1}}}+\underbrace{\frac{1/4}{1+z^{-1}}}$$

$$\boxed{=\frac{1}{2}(n+1)\,u(n+1)+\frac{1}{4}u(n)+\frac{1}{4}(-1)^n u(n)}$$

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Örnek! $X(z) = \dfrac{1+\frac{1}{4}z^{-1}}{\left(1-\frac{1}{2}z^{-1}\right)^2}$ $|z| > \frac{1}{2}$

$$= \frac{A}{\left(1-\frac{1}{2}z^{-1}\right)^2} + \frac{B}{\left(1-\frac{1}{2}z^{-1}\right)}$$

$$A = \frac{\cancel{\left(1-\frac{1}{2}z^{-1}\right)^2} \cdot \left(1+\frac{1}{4}z^{-1}\right)}{\cancel{\left(1-\frac{1}{2}z^{-1}\right)^2}}\Bigg|_{z^{-1}=2} = \frac{3}{2}$$

$$B = \frac{d}{dz}A(z) = \frac{d}{dz}\left(1+\frac{1}{4}z^{-1}\right)\Bigg|_{z^{-1}=2} = 0 - \frac{1}{4}z^{2}\Bigg|_{z^{-1}=2} = -1$$

$$= \frac{3/2}{\left(1-\frac{1}{2}z^{-1}\right)^2} - \overbrace{\frac{1}{\left(1-\frac{1}{2}z^{-1}\right)}}^{\to \left(\frac{1}{2}\right)^n u(n)}$$

$$\frac{3/2}{\left(1-\frac{1}{2}z^{-1}\right)^2} = c\frac{d}{dz}\left(\frac{3/2}{1-\frac{1}{2}z^{-1}}\right) = c.\frac{3}{2}.\frac{\left(0.\cancel{\left(1-\frac{1}{2}z^{-1}\right)}\right) - 1\left(0+\frac{1}{2}z^{-2}\right)}{\left(1-\frac{1}{2}z^{-1}\right)^2}$$

$$= \left(\frac{1}{2}\right)^{n+1} u(n+1).(n+1).3$$

$$= 3(n+1)\left(\frac{1}{2}\right)^{n+1}.u(n+1) - \left(\frac{1}{2}\right)^n u(n)$$

Örnek!

$$X(z)=\frac{4-\frac{7}{4}z^{-1}+\frac{1}{4}z^{-2}}{\left(1-\frac{1}{2}z^{-1}\right)\left(1-\frac{1}{4}z^{-1}\right)}=\frac{4-\frac{7}{4}z^{-1}+\frac{1}{4}z^{-2}}{1-\frac{1}{4}z^{-1}-\frac{1}{2}z^{-1}+\frac{1}{8}z^{-2}}$$

$$\begin{array}{l|l} \cancel{\frac{1}{4}z^{-2}}-\frac{7}{4}z^{-1}+4 & \frac{1}{8}z^{-2}-\frac{3}{4}z^{-1}+1 \\ \hline -\left(\cancel{\frac{1}{4}z^{-2}}-\frac{6}{4}z^{-1}+2\right) & 2 \\ \hline -\frac{1}{4}z^{-1}+2 & \end{array}$$

$$=2+\frac{2-\frac{1}{4}z^{-1}}{\left(1-\frac{1}{2}z^{-1}\right)\left(1-\frac{1}{4}z^{-1}\right)}=2+\frac{A}{1-\frac{1}{2}z^{-1}}+\frac{B}{1-\frac{1}{4}z^{-1}}$$

$$A=\left.\frac{2-\frac{1}{4}z^{-1}}{1-\frac{1}{4}z^{-1}}\right|_{z^{-1}=2}\quad\frac{2-\frac{1}{4}\cdot 2}{1-\frac{1}{4}\cdot 2}=\frac{2-\frac{1}{2}}{1-\frac{1}{2}}=\underline{\underline{3}}$$

$$B=\left.\frac{2-\frac{1}{4}z^{-1}}{1-\frac{1}{2}z^{-1}}\right|_{z^{-1}=4}=\frac{2-\frac{1}{\cancel{4}}\cdot\cancel{4}}{1-\frac{1}{2}\cdot 4}=\frac{1}{-1}=-\underline{1}$$

$$=2\cdot 1+\frac{3}{1-\frac{1}{2}z^{-1}}-\frac{1}{1-\frac{1}{4}z^{-1}}\Rightarrow 2\delta(n)+3\left(\frac{1}{2}\right)^{n}u(n)-\left(\frac{1}{4}\right)^{n}u(n)$$

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Örnek! $X(z) = \dfrac{z^{-2}+2z^{-1}+2}{z^{-1}+1}$ $\quad |z|>1$

$$\begin{array}{r|l} z^{-2}+2z^{-1}+2 & z^{-1}+1 \\ \cline{2-2} -\;\; z^{-2}+z^{-1} & z^{-1}+1 \\ \cline{1-1} z^{-1}+2 & \\ -\;\; z^{-1}+1 & \\ \cline{1-1} 1 & \end{array}$$

$$X(z) = \frac{(z^{-1}+1)(z^{-1}+1)+1}{(z^{-1}+1)} = z^{-1}+1+\frac{1}{z^{-1}+1}$$

$$= \delta(n-1)+\delta(n)-(-1)^n u(-n-1)$$

Örnek! $X(z) = \dfrac{z^{-2}+z^{-1}}{(1-z^{-1})(1+3z^{-1})}$

$$\begin{array}{r|l} z^{-2}+z^{-1} & -3z^{-2}+2z^{-1}+1 \\ \cline{2-2} -\;\; z^{-2}-\frac{2}{3}z^{-1}-\frac{1}{3} & -\frac{1}{3} \\ \cline{1-1} \frac{5}{3}z^{-1}+\frac{1}{3} & \end{array}$$

$$X(z) = \frac{(-3z^{-2}+2z^{-1}+1)\left(\frac{1}{3}\right)+\left(\frac{5}{3}z^{-1}+\frac{1}{3}\right)}{\cancel{----}}$$

$$-\frac{1}{3}+\frac{\left(\frac{5}{3}z^{-1}+\frac{1}{3}\right)}{(1-z^{-1})(1+3z^{-1})} = -\frac{1}{3}+\frac{A}{(1-z^{-1})}+\frac{B}{1+3z^{-1}}$$

$$A = \left.\frac{\frac{5}{3}z^{-1}+\frac{1}{3}}{1+3z^{-1}}\right|_{z^{-1}=1} = \frac{1}{2}$$

$$B = \left.\frac{\frac{5}{3}z^{-1}+\frac{1}{3}}{1-z^{-1}}\right|_{z^{-1}=-\frac{1}{3}} = -\frac{1}{6}$$

$$= -\frac{1}{3}+\frac{1/2}{1-z^{-1}}-\frac{1/6}{1+3z^{-1}}$$

$$-\frac{1}{3}\delta(n)+\frac{1}{2}u(n)-\frac{1}{6}(-3)^n u(n)$$

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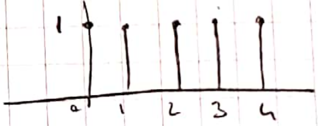
Örnek: $X(z) = \dfrac{1-z^{-5}}{1-z^{-1}} \quad |z|>1$

$$\begin{array}{r|l} -z^{-5}+1 & -z^{-1}+1 \\ \hline & z^{-4}+z^{-3}+z^{-2}+z^{-1}+1 \\ \overline{\qquad\quad} & \\ -z^{-1}+1 & \end{array}$$

$$= \frac{(\cancel{-z^{-1}+1})(z^{-4}+z^{-\cancel{2}3}+z^{-2}+z^{-1}+1)}{\cancel{1-z^{-1}}} = u(n)-u(n-5)$$

$=$



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4.11.2019

Fark denklemleri:

Örnek: $y(n)-3y(n-1)-4y(n-2)=x(n)+2x(n-1)$

$x(n)=4^n\cdot u(n)$

$y(n)=?$

$$y(z)-3z^{-1}y(z)-4z^{-2}y(z)=x(z)+2z^{-1}\cdot x(z)$$

$$y(z)(1-3z^{-1}-4z^{-2})=x(z)\cdot(1+2z^{-1})$$

$$y(z)=\frac{1+2z^{-1}}{1-3z^{-1}-4z^{-2}}\cdot x(z)=\frac{1+2z^{-1}}{(1-3z^{-1}-4z^{-2})(1-4z^{-1})}$$

$$\begin{matrix}1 & -4\\ 1 & 1\end{matrix}$$

$$=\frac{1+2z^{-1}}{(1+z^{-1})(1-4z^{-1})(1-4z^{-1})}=\frac{1+2z^{-1}}{(1+z^{-1})(1-4z^{-1})^2}=\frac{A}{(1-4z^{-1})^2}+\frac{B}{1-4z^{-1}}+\frac{C}{1+z^{-1}}$$

$$A=\cancel{(1-4z^{-1})^2}\frac{1+2z^{-1}}{(1+z^{-1})\cancel{(1-4z^{-1})^2}}\Bigg|_{z^{-1}=\frac{1}{4}}=\frac{1+2\cdot\frac{1}{4}}{1+\frac{1}{4}}=\frac{6}{5}$$

$$B=\frac{d}{dz}A(z)=\frac{d}{dz}\cdot\frac{1+2z^{-1}}{1+z^{-1}}\Bigg|_{z^{-1}=\frac{1}{4}}\frac{(0-2z^{-1})(1+z^{-1})-(1+2z^{-1})\cdot(-z^{-2})}{(1+z^{-1})^2}=-\frac{1}{25}$$

$$=\frac{6/5}{(1-4z^{-1})^2}-\frac{1/25}{1-4z^{-1}}-\frac{1/25}{1+z^{-1}}$$

$$C=\frac{(1+z^{-1})(1+2z^{-1})}{(1+z^{-1})(1-4z^{-1})^2}\Bigg|_{z^{-1}=-1}=-\frac{1}{25}$$

$$=\frac{6/5}{(1-4z^{-1})^2}=K\frac{d}{dz}\frac{6/5}{1-4z^{-1}}$$

$$K=-\frac{1}{4}z^2$$

$$=\frac{1}{4}\cdot z\cdot n\cdot\frac{6}{5}\cdot 4^n\,u(n)$$

$$=\frac{6}{20}\cdot 4^{n+1}(n+1)\cdot u(n+1)+\dots$$

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Fourier açılımı

$t \to a_k$

$$x(t) = \sum_{-\infty}^{\infty} a_k \cdot e^{jk w_0 t} \qquad a_k = \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t) \cdot e^{-jwt} \cdot dt$$

$e^{jkw_0t} \downarrow z^{-n}$

w_0 = Temel frekans

$$w_0 = \frac{2\pi}{T_0}$$

$k = \mp 1$ 1. harmonik

$k = \mp 2$ 2. harmonik

$$\boxed{e^{jt} = \cos\theta + j \sin\theta}$$

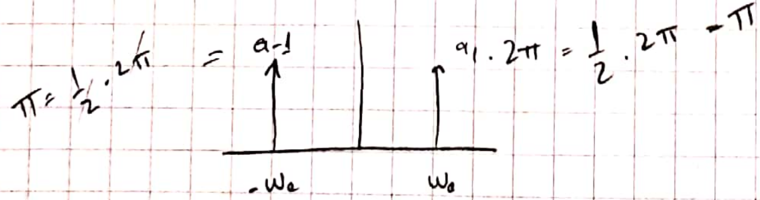
$$\boxed{\cos\theta = \frac{e^{jt} + e^{-jt}}{2}} \qquad \boxed{\sin\theta = \frac{e^{jt} - e^{-jt}}{2j}}$$

$$\boxed{\sin^2\theta + \cos^2\theta = 1} \qquad \boxed{\cos^2\theta - \sin^2\theta = \cos 2\theta}$$

Örnek!

$$x(t) = \frac{e^{jw_0t} + e^{-jw_0t}}{2} = \frac{1}{2} e^{\overbrace{jw_0 t}^{k=1}} + \frac{1}{2} e^{\overbrace{-jw_0 t}^{k=-1}}$$

$$a_1 = \frac{1}{2} \qquad a_{-1} = \frac{1}{2}$$



Örnek! $x(t) = \sin\left(2t + \frac{\pi}{6}\right)$

$$= \frac{e^{j(2t+\frac{\pi}{6})} - e^{-j(2t+\frac{\pi}{6})}}{2j}$$

$$= \frac{e^{j2t} \cdot e^{j\frac{\pi}{2}} - e^{-2jt} \cdot e^{-j\frac{\pi}{6}}}{2j}$$

$$e^{j\frac{\pi}{6}} = \cos\frac{\pi}{6} + j\sin\frac{\pi}{6}$$

$$= \cos 30^\circ + j\sin 30^\circ$$

$$= \frac{\sqrt{3}}{2} + \frac{1}{2}$$

$$= \frac{\frac{\sqrt{3}}{2} + \frac{j}{2}}{2j} \underbrace{e^{j2t}}_{k=1} - \frac{\frac{\sqrt{3}}{2} - \frac{j}{2}}{2j} \underbrace{e^{-j2t}}_{k=-1}$$

$$= \left(\frac{\sqrt{3}}{4j} + \frac{1}{4}\right) = a_1$$

$$= \left(\frac{\sqrt{3}}{4j} - \frac{1}{4}\right) = a_{-1}$$

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Örnek: $x(t) = \sin\frac{t}{4} + \cos t + \cos\frac{t}{2}$

a) temel frekansını bul

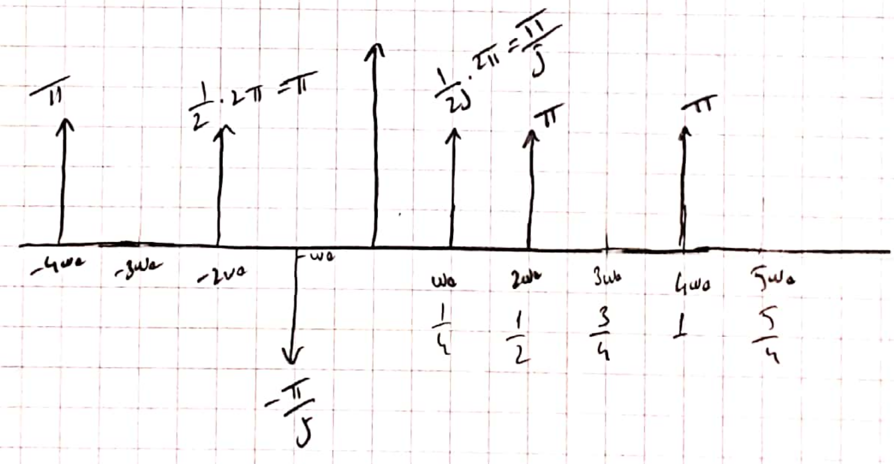
b) seri katsayısını bul

c) frekans spektrumunu çiz

a) EBOB $\left(\frac{1}{4}, 1, \frac{1}{2}\right) = \frac{1}{4}$

b) $$x(t) = \frac{e^{jw_0t} - e^{-jw_0t}}{2j} + \frac{e^{jw_0 4t} + e^{-jw_0 4t}}{2} + \frac{e^{jw_0 2t} + e^{-jw_0 2t}}{2}$$

$\boxed{a_1 = \frac{1}{2j}}$ $\quad a_{-1} = -\frac{1}{2j} \quad$ $a_2 = \frac{1}{2} = a_{-2} \quad$ $a_4 = \frac{1}{2} = a_{-4}$



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Örnek ! $x(t) = \sin^2 2t \quad = \frac{1}{2} - \frac{1}{2}\cos 4t \quad w_0 = 4$

$$\cancel{\cos^2 2t} + \sin^2 2t = 1$$

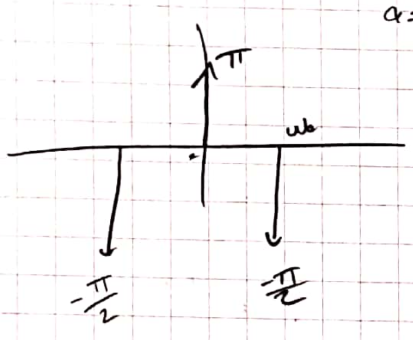
$$- \cancel{\cos^2 2t} + \sin^2 2t = -\cos 4t$$

$$2\sin^2 2t = 1 - \cos 4t$$

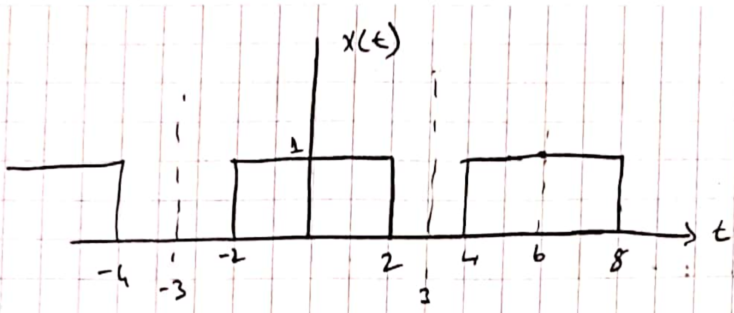
$$\sin^2 2t = \frac{1 - \cos 4t}{2}$$

$$= \frac{1}{2} - \frac{1}{2}\left(\frac{e^{j w_0 t} + e^{-j w_0 t}}{2}\right)$$

$$= \underbrace{\frac{1}{2}}_{a = \frac{1}{2}} e^{j w_0 t} \underbrace{- \frac{1}{4}}_{a_1 = -\frac{1}{4}} e^{j w_0 t} \underbrace{- \frac{1}{4}}_{a_{-1} = -\frac{1}{4}} e^{j w_0 t}$$



Örnek !



bu sinyali fourier serisine açın. yani $a_k = ?$

$$a_k = \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{T_0} x(t).e^{-jw_0 t}.dt = \frac{1}{6}\int_{-3}^{3} x(t)\, e^{-jw_0 t}.dt$$

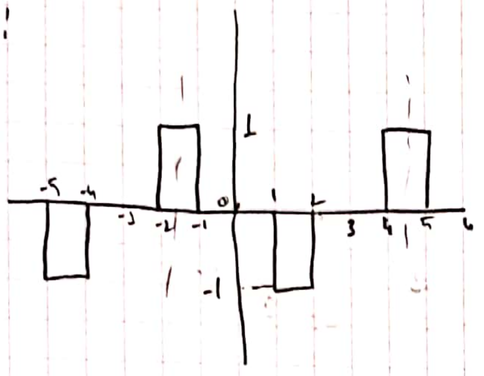
$$= \frac{1}{6}\int_{-2}^{2} 1.e^{-jw_0 t}.dt =$$

$$= \frac{1}{6}\left(-\frac{1}{jw_0}.e^{-jw_0 t}\Big|_{-2}^{2}\right) = -\frac{1}{6jw_0}\left(e^{-jw_0.2k} - e^{2jw_0 t}\right)$$

$$= \frac{1}{6jw_0 k}.e^{2jw_0 k} - \frac{1}{6jw_0 k}e^{-2jw_0 k} = \frac{1}{3w_0 k}\left(\frac{e^{2jw_0 k} - e^{2jw_0 k}}{2j}\right)$$

$$= \frac{1}{3w_0 k}.\sin 2w_0 k = \frac{1}{\pi k}.\sin\frac{2\pi}{3}.k$$

Örnek:



$a_k = ?$

$T_0 = 6$

$\omega_0 = \dfrac{2\pi}{6} = \dfrac{\pi}{3}$

$$a_k = \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t)\cdot e^{-jk\omega_0 t}\cdot dt = \frac{1}{6}\int_{-3}^{3} x(t)\cdot e^{jk\omega_0 t}\cdot dt$$

$$= \frac{1}{6}\left[\int_{-2}^{-1} 1\, e^{-jk\omega_0 t}\cdot dt + \int_{1}^{2}(-1)\, e^{-jk\omega_0 t}\cdot dt\right]$$

$$= \frac{1}{6}\left(-\frac{1}{jk\omega_0} e^{-jk\omega_0 t}\Big|_{-2}^{-1} + \frac{1}{jk\omega_0} e^{-jk\omega_0 t}\Big|_{1}^{2}\right)$$

$$= \frac{1}{6jk\omega}\left[\left(e^{-jk\omega_0 2} - e^{-jk\omega_0 1}\right) - \left(e^{jk\omega_0} - e^{jk\omega_0 2}\right)\right]$$

$$= \left(e^{jk\omega_0 -2} + e^{-jk\omega_0 2} - e^{jk\omega_0} - e^{-jk\omega_0}\right)$$

$$= \frac{1}{3k\omega_0}\left(\cos 2k\omega_0 - \cos k\omega_0\right) = \frac{1}{jk\pi}\left(\cos\frac{2\pi k}{3} - \cos\frac{\pi}{3}k\right)$$

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11.12.2019

Fourier Açılımı özellikleri:

- Doğrusallık

$x(t) \rightarrow a_k$

$y(t) \rightarrow b_k$ $\quad\quad z(t) = a \cdot x(t) + b \cdot y(t)$

$z(t) \rightarrow c_k$ $\quad\quad \boxed{c_k = a \cdot a_k + b \cdot b_k}$

- Öteleme

$x(t-t_0) \rightarrow e^{-jw_0 t_0} \cdot a_k$

- Türev

$\frac{d}{dt} x(t) \rightarrow jk w_0 \cdot a_k$

Fourier Dönüşümü

$$X(w) \int_{-\infty}^{\infty} x(t) \cdot e^{-jwt} \cdot dt$$

Ters Fourier Dönüşümü

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(w) \cdot e^{jwt} \cdot dw$$

Örnek : $x(t) = e^{-at} . u(t)$ $\quad X(w) = ?$

$$X(w) = \int_{-\infty}^{\infty} e^{-at} . u(t) . e^{-jwt} . dt$$

$$= \int_{0}^{\infty} e^{-at} . 1 . e^{-jwt} . dt = \int_{0}^{\infty} e^{-t(a+jw)} . dt$$

$$= \frac{1}{a+jw} . e^{-t(a+jw)} \Big|_{0}^{\infty} = -\frac{1}{a+jw} (0-1) = \frac{1}{a+jw}$$

Örnek : $x(t) = \delta(t)$ $\quad X(w) = ?$

$$X(w) = \int_{-\infty}^{\infty} \delta(t) . e^{-jwt} . dt = \int_{t=0} \delta(t) . e^{-jwt} . dt$$

$$= \underbrace{\delta(0)}_{1} . e^{-jw.0} = 1$$

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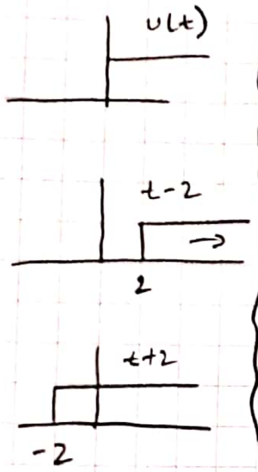
Örnek! $x(t) = 4u(t-2) + 2u(t) - 4u(t+2)$ $\qquad X(w) = ?$

$$X(w) = 4\int u(t-2).e^{-jwt}.dt + 2\int u(t).e^{-jwt}dt - 4\int u(t+2).e^{-jwt}dt$$



$$= -\frac{4}{jw}.e^{-jwt}\Big|_{2}^{\infty} - \frac{2}{jw}e^{-jwt}\Big|_{0}^{\infty} + \frac{4}{jw}e^{-jwt}\Big|_{-2}^{\infty}$$

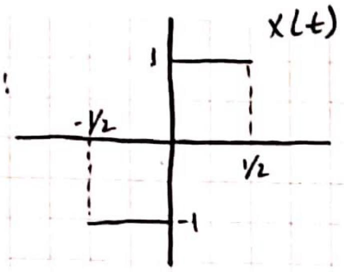
$$= -\frac{4}{jw}\left(0 - e^{-2jw}\right) - \frac{2}{jw}(0-1) + \frac{4}{jw}\left(0 - e^{2jw}\right)$$

$$= \frac{4}{jw}e^{-2jw} + \frac{2}{jw} - \frac{4}{jw}e^{2jw}$$

$$= \frac{2}{jw} - \frac{4}{jw}\left(e^{2jw} - e^{-2jw}\right)$$

$$X(w) = \frac{2}{jw} - \frac{8}{w}\sin 2w$$

<u>örnek</u>:



$X(\omega) = ?$

$$X(\omega) = \int_{-1/2}^{1/2} x(t) . e^{-j\omega t} . dt$$

$$= \int_{-1/2}^{0} (-1) . e^{-j\omega t} . dt + \int_{0}^{\frac{1}{2}} 1 . e^{-j\omega t} . dt$$

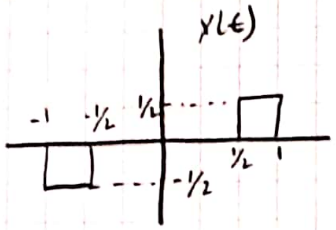
$$= + \frac{1}{j\omega} . e^{-j\omega t} \Big|_{-1/2}^{0} - \frac{1}{j\omega} e^{-j\omega t} \Big|_{0}^{1/2}$$

$$= \frac{1}{j\omega} \left(1 - e^{\frac{j\omega}{2}} \right) - \frac{1}{j\omega} \left(e^{-\frac{j\omega}{2}} - 1 \right)$$

$$= \frac{2}{j\omega} - \frac{1}{j\omega} \left(e^{\frac{j\omega}{2}} + e^{-\frac{j\omega}{2}} \right) = \frac{2}{j\omega} - \frac{2}{j\omega} . \cos \frac{\omega}{2}$$

Tarih:........../........../..........

Örnek!



$X(\omega) = ?$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} \cdot dt$$

$$X(\omega) = \int_{-1/2}^{1} \left(\frac{1}{2}\right) e^{-j\omega t} \cdot dt + \int_{-1}^{-1/2} \left(-\tfrac{1}{2}\right) \cdot e^{-j\omega t} \cdot dt$$

$$= -\frac{1}{2j\omega} e^{-j\omega t} \Big|_{1/2}^{1} + \frac{1}{2j\omega} e^{-j\omega t} \Big|_{-1}^{-1/2}$$

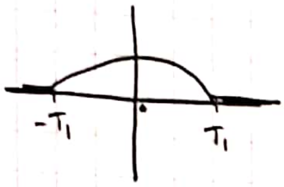
$$= -\frac{1}{2j\omega} \left(e^{-j\omega} - e^{-\frac{j\omega}{2}}\right) + \frac{1}{2j\omega} \left(e^{\frac{j\omega}{2}} - e^{j\omega}\right)$$

$$= -\frac{1}{2j\omega} \left(e^{j\omega} + e^{-j\omega}\right) + \frac{1}{2j\omega} \left(e^{\frac{j\omega}{2}} + e^{-\frac{j\omega}{2}}\right)$$

$$= \frac{1}{j\omega} \cos\frac{\omega}{2} - \frac{1}{j\omega} \cos\omega$$

Örnek !

$$x(t) = \begin{cases} 0 & |t| > T_1 \\ \cos(\pi . t) & |t| < T_1 \end{cases}$$



$$X(\omega) = \int_{-T_1}^{T_1} \cos(\pi . t)\, e^{-j\omega t} . dt$$

$$= \int_{-T_1}^{T_1} \left(\frac{e^{j\pi t} + e^{-j\pi t}}{2} \right) . e^{-j\omega t} . dt$$

$$= \frac{1}{2} \int e^{jt(\pi - \omega)} dt + \frac{1}{2} \int e^{-jt(\pi + \omega)} dt$$

$$= \frac{1}{2j(\pi-\omega)} . e^{jt(\pi-\omega)} \Big|_{-T_1}^{T_1} - \frac{1}{2j(\pi-\omega)} . e^{-jt(\pi-\omega)} \Big|_{-T_1}^{T_1}$$

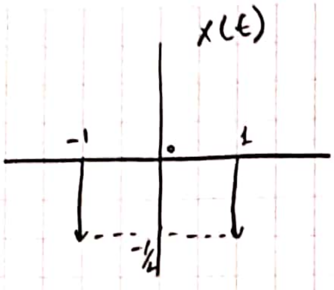
$$= \frac{1}{2j(\pi-\omega)} \left(\frac{e^{jT_1(\pi-\omega)} - e^{-jT_1(\pi-\omega)}}{} \right) - \frac{1}{2j(\pi-\omega)} \left(\frac{e^{-jT_1(\pi-\omega)} - e^{-jT_1(\pi-\omega)}}{} \right)$$

$$= \frac{1}{\pi - \omega} \sin(\pi - \omega) T_1 - \frac{1}{\pi + \omega} \sin(\pi + \omega) T_1$$

Pegi Teknik

Tarih:............/............./............

örnek :



$X(\omega) = ?$

$$x(t) = -\frac{1}{2}\,\delta(t-1) + \delta(t) - \frac{1}{2}\,\delta(t+1)$$

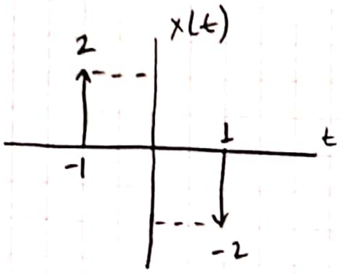
$$X(\omega) = \int_{-\infty}^{\infty} x(t)\,.\,e^{-j\omega t}\,dt - \frac{1}{2}\int_{t=1} \delta(t-1)\,.\,e^{-j\omega t}\,.\,dt + \int_{t=0} \delta(t)\,.\,e^{-j\omega t} - \frac{1}{2}\int_{t=-1} \delta(t+1)\,.\,e^{-j\omega t}\,.\,dt$$

$$= \frac{1}{2}\left(e^{-j\omega}\right) + 1 - \frac{1}{2}\,e^{j\omega} = 1 - \frac{1}{2}\left(e^{j\omega} + e^{-j\omega}\right)$$

$$= \boxed{1 - \cos\omega}$$

Tarih: ..../..../....

Örnek !



$X(\omega) = ?$

$x(t) = -2\delta(t-1) + 2\delta(t+1)$

$$X(\omega) = -2 \int_{t=1} \delta(t-1) \cdot e^{-j\omega t} \, dt + 2 \int_{t=-1} \delta(t+1) e^{-j\omega t} \, dt$$

$$= -2e^{-j\omega} + 2e^{j\omega} = 2(e^{j\omega} - e^{-j\omega})$$

$$= 4j - \sin\omega$$

Fourier Dönüşümü Özellikleri

- Doğrusallık

$$z(t) = a \cdot x(t) + b \cdot y(t)$$

$$Z(\omega) = a \cdot X(\omega) + b \cdot Y(\omega)$$

- Öteleme

$$x(t - t_0) \xrightarrow{F} e^{-j\omega t_0} \cdot X(\omega)$$

Türev

$$\frac{d}{dt} x(t) \xrightarrow{F} j\omega \cdot X(\omega)$$

- Ölçekleme

$$x(at) \xrightarrow{F} \frac{1}{|a|} X\left(\frac{\omega}{a}\right)$$

- Ters çevirme

$$x(t) \rightarrow X(\omega)$$

$$X(t) \rightarrow 2\pi \cdot x(-\omega)$$

- Çarpma

$$r(t) = a(t) \cdot b(t)$$

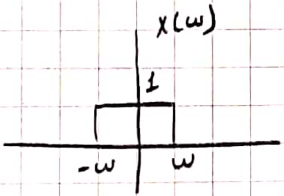
$$R(w) = \frac{1}{2\pi}\left[A(w) * B(w)\right]$$

- (İsimsiz özellik)

$$x(t) \longrightarrow X(w)$$

$$t \cdot x(t) \longrightarrow j\frac{d}{dw} X(w)$$

Örnek !



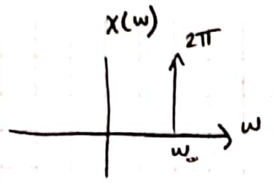
$x(t) = ?$

$$x(t) = \frac{1}{2\pi}\int_{-w}^{w} 1 \cdot e^{jwt}\, dw = \frac{1}{2\pi j t} \cdot e^{jwt}\Big|_{-w}^{w}$$

$$= \frac{1}{2\pi j t}\left(e^{jwt} - e^{-jwt}\right)$$

$$= \frac{1}{\pi t} \sin wt$$

Örnek!



$x(t) = ?$

$$X(w) = 2\pi\,\delta(w - w_0)$$

$$x(t) = \frac{1}{2\pi} \int_{w_0} \cancel{2\pi}\, \underbrace{\delta(w - w_0)}_{1} . e^{jwt} . dw$$

$$= \underline{e^{jw_0 t}}$$

Örnek! $\dfrac{dy(t)}{dt} + a y(t) = x(t)$ ise $h(t) = ?$

$$jw . Y(w) + a . Y(w) = X(w)$$

$$Y(w) . [jw + a] = X(w)$$

$$\frac{Y(w)}{X(w)} = h(w) = \frac{1}{a + jw} \Rightarrow \boxed{h(t) = e^{-at} . u(t)}$$

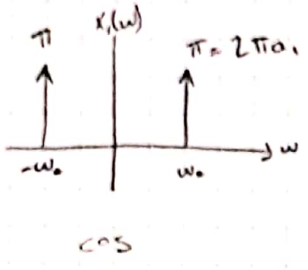
Örnek! $\ddot{y}(t) + \dot{y}(t) + 3y(t) = \dot{x}(t) + x(t)$

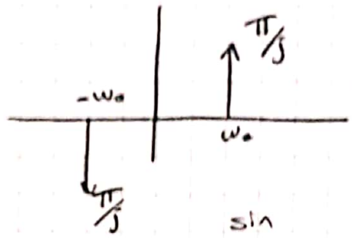
$$(jw)^2 Y(w) + 4jw\,Y(w) + 3\,Y(w) = jw . X(w) + X(w)$$

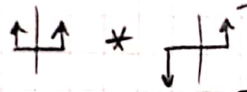
$$Y(w)\left[(jw)^2 + 4(jw) + 3\right] = X(w)\,[jw + 1]$$

$$\frac{Y(w)}{X(w)} = \frac{\cancel{1 + jw}}{\cancel{(jw+1)}\,(jw+3)} = \frac{1}{3 + jw} \Rightarrow \underline{h(t) = e^{-3t} . u(t)}$$

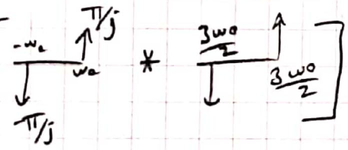
Örnek! $x(t) = \underbrace{\cos w_0 t}_{x_1(t)} \, . \, \underbrace{\sin w_0 t}_{x_2(t)}$

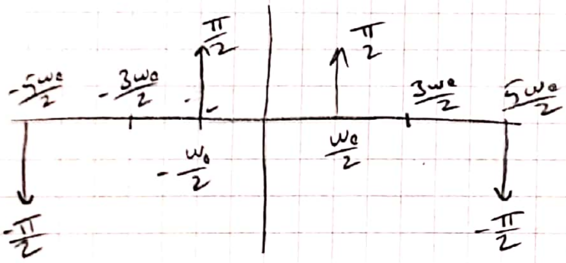




$X(\omega) = \frac{1}{2\pi}$ [] = - - - -

Örnek! $x(t) = \sin w_0 t \, . \, \sin \frac{3 w_0 t}{2}$ $\quad \boxed{j^2 = -1}$

$X(\omega) = \frac{1}{2\pi}$ [] $= \frac{1}{\cancel{2\pi}} \cdot \frac{\cancel{\pi}}{j} \cdot \frac{\pi}{j} = \frac{\pi}{2j^2} = -\frac{\pi}{2}$



$X(\omega) = -\frac{\pi}{2} \delta\left(w - \frac{5w_0}{2}\right) + \frac{\pi}{2} \delta(w - w_0)$ - - -

$$x(t) = \frac{1}{2} \, . \cos \frac{w_0 t}{2} - \frac{1}{2} \cos \frac{5 w_0 t}{2}$$

Örnek! $X(\omega) = \frac{\pi}{2j}\left[\delta\left(w + \frac{3\pi}{2}\right) - \delta\left(w - \frac{3\pi}{2}\right) - \frac{2\pi}{3}\left[\delta(w-\pi) + \delta(w+\pi)\right]\right]$

$\text{EBOB}\left(3\frac{\pi}{2}, \pi\right) = \frac{\pi}{2} = w_0$

$X(\omega) = \frac{\pi}{2j}\left[\delta(w + 3w_0) - \delta(w - 3w_0)\right] - \frac{2\pi}{3}\left[\delta(w - 2w_0) + \delta(w + 2w_0)\right]$

$= \frac{1}{\cancel{2\pi}} \cdot \frac{\cancel{\pi}}{2j} e^{-3jw_0 t} - \frac{1}{2\pi} \cdot \frac{\pi}{2j} e^{3jw_0 t} - \frac{2\pi}{3} \cdot \frac{1}{2\pi} e^{j2w_0 t} - \frac{2\pi}{3} \cdot \frac{1}{2\pi} e^{-2jw_0 t}$

$$= \frac{1}{4j} \cdot e^{-3j\omega_0 t} - \frac{1}{4j} e^{3j\omega_0 t} - \frac{1}{3} e^{2j\omega_0 t} - \frac{1}{3} e^{-2j\omega_0 t}$$

a) $\omega_0 = \frac{\pi}{2}$

b) Seri katsayıları:

$a_{-3} = \frac{1}{4j}$ $a_3 = -\frac{1}{4j}$ $a_2 = a_{-2} = \frac{1}{3}$

c) Zaman domeni

$$\frac{1}{2}\sin 3\omega_0 t - \frac{2}{3}\cos 2\omega t$$

End of 11.12.2019

18.12.2019

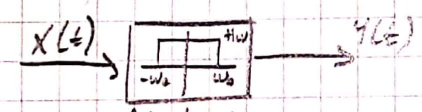
1 - Ters z

2 - Fourier serisi

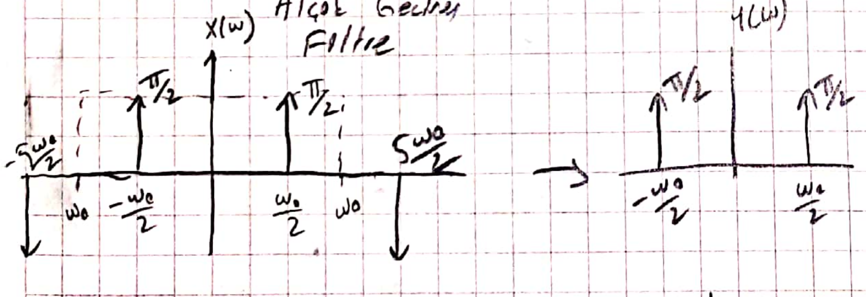
3 - " dönüşümü veya Ters fourier dönüşümü

4 - örnekler

Örnek: $x(t) = \frac{1}{2}\left(\cos\frac{\omega_0}{2}t - \cos\frac{5\omega_0}{2}t\right)$



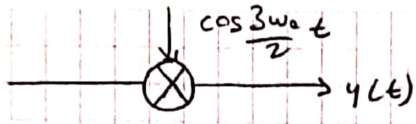
Alçak Geçiren Filtre

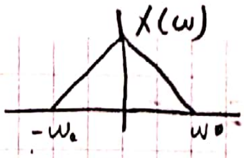


$$y(t) = \frac{1}{2}\cos\frac{\omega_0}{2}t$$

cos

Örnek: $x(t)$ $\cos \frac{3w_0}{2} t$ $y(t)$

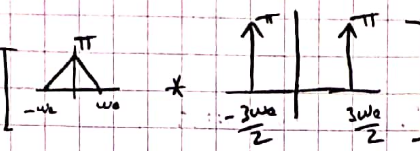


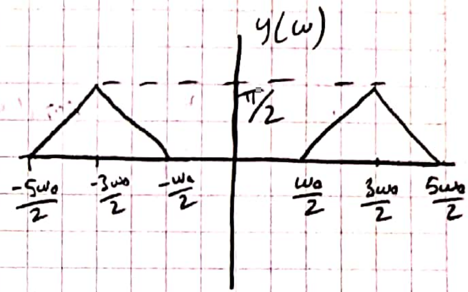


$y(w)$ çizimiz?

$$y(t) = x(t) \,.\, \cos \frac{3w_0}{2} . t$$

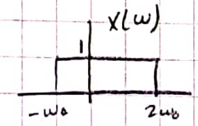
$$Y(w) = \frac{1}{2\pi} \left[\quad * \quad \right] =$$

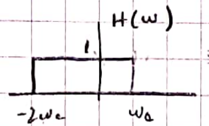




$$= \frac{1}{2\cancel{\pi}} . \cancel{\pi} . \pi = \frac{\pi}{2}$$

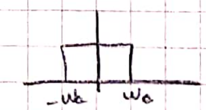
Örnek:





$y(t) = ?$

$$Y(w) = X(w) . H(w) \longrightarrow$$



$$x(t) = \frac{1}{2\pi} \int_{-w_0}^{w_0} Y(w) . e^{jwt} . dw = \frac{1}{2\pi} . \frac{1}{jt} . e^{jwt} \Big|_{-w_0}^{w_0} = \frac{1}{2\pi j t} . \left(e^{jw_0 t} - e^{-jw_0 t} \right)$$

$$= \frac{\sin w_0 t}{\pi t}$$

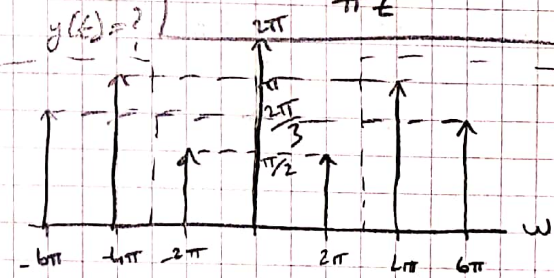
Örnek: $w_0 = 2\pi$ $x(t)$ $a_0 = 1$

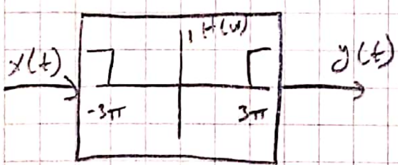
$y(t) = ?$

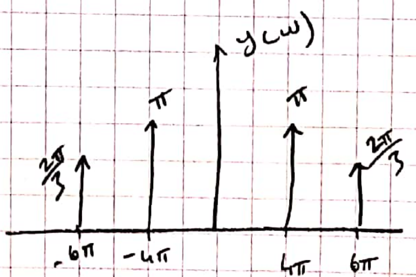
$a_1 = a_{-1} = \frac{1}{4}$

$a_2 = a_{-2} = \frac{1}{2}$

$a_3 = a_{-3} = \frac{1}{3}$





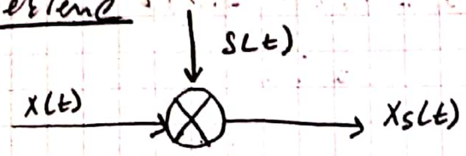


$$\boxed{y(t) = \cos 4\pi + \frac{2}{3} \cos 6\pi}$$

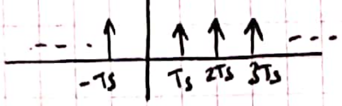
Pegi Teknik

Tarih:........../........../..........

<u>Örnekleme</u>:



$$s(t) = \sum_{-\infty}^{\infty} \delta(t - kT_s)$$



$$w(s) = \frac{2\pi}{T_s} = 2\pi f_s$$

↓ örnekleme frekansı

$x[n] \Rightarrow t \to n.T_s$

$x(n) = x(n+N)$

periyodiklik şartı:

$$N = \frac{2\pi k}{w_0 T_s}$$

<u>Örnek</u>: $x(t) = e^{jw_0 t}$ ise $x(n) = ?$

$t \to n.T_s$ $\qquad x(n) = e^{jw_0 . n . T_s}$

b) periyodikliği →

$$x(n) = e^{jw_0 n . T_s} = e^{jw_0 (n+N) T_s}$$

$$= \cancel{e^{jw_0 n T_s}} = \cancel{e^{jw_0 n T_s}} . e^{jw_0 N T_s}$$

$$= e^{jw_0 N T_s} = 1 = e^{j2\pi k}$$

$$w_0 N T_s = 2\pi . k \qquad N = \frac{2\pi k}{w_0 T_s}$$

Örnek : $x(t) = x_a(t) = \cos 15t$ $T_s = 0{,}1\pi$ saniye ise

$x(n)$'i ve periyodik olup olmadığını bulun.

$t \to n.T_s$

$x(n) = \cos 15.n.T_s = \cos 15n.\frac{1\pi}{10} = \cos \frac{3\pi}{2} n$

$w_0 = 15$

$N = \frac{2\pi k}{w_0 T_s} = \frac{2\pi k}{w_0 . \frac{1}{100}\pi} = \frac{2k}{\frac{3}{2}} = \frac{4}{3} k$

k ve N tamsayı bu yüzden periyodik

$k = 3$ için
$N = 4$

Örnek : $f_s = 10k\,Hz$

$x(n) = \sin\left(\frac{n\pi}{4}\right)$

$t \to nT_s$ veya $t \to \frac{n}{f_s}$ kullanılabilir.

$x(t) = ?$

$x(n) = \sin\left(\frac{n\pi}{4}\right) = \sin\left(\frac{n}{10.000} . 10.000 . \frac{\pi}{4}\right)$

$nT_s = t$

$x(t) = \sin 2500\pi t$

$w_0 = 2500\pi$

Pegi Teknik

Tarih:........../........../..........

Örnek: $x(t) = \sin 2500\pi t$

$f_s = 2{,}5\ kHz$

$x(n) = ?$

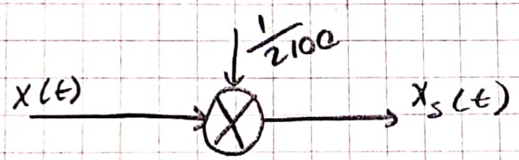
1. $T_s = \frac{1}{f_s} = \frac{1}{2500}$ $\qquad x(n) = \sin 2500 \cdot \pi \cdot \frac{1}{2500}$

$= \sin \pi n$

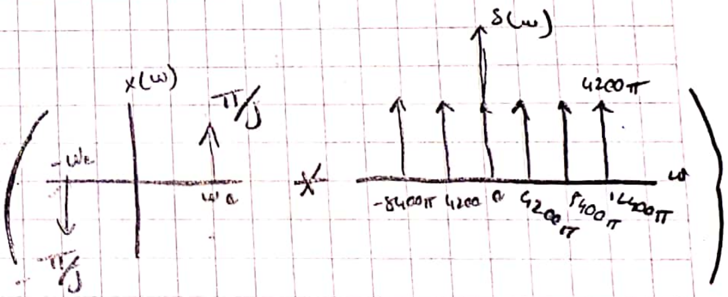
Örnek: $f_0 = 1\ kHz$

$x(t) = \sin w_0 t$ $\qquad$ Örneklenmiş işaretin frekans spektrumunu çizin

$T_s = \frac{1}{2100}$ sn



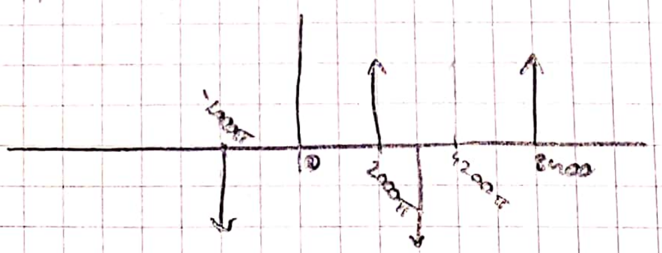
$X_s(w) = \frac{1}{2\pi}\left[X(w) * \delta(w)\right] = \frac{1}{2\pi}\Big($



$\Big)$

$w_s = \frac{2\pi}{T_s} = \boxed{4200\pi}$

$w_0 = 2\pi f_0 = 2000\pi$

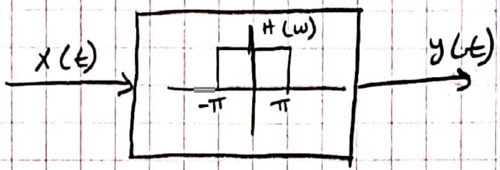


$X(s) = \frac{1}{2\pi} \cdot \frac{\pi}{j} \cdot 4200\pi = 2100\frac{\pi}{j}$

2019 Final Sorusu!

$x(t) = \cos \frac{2\pi}{3} t$

$T_s = 2sn$



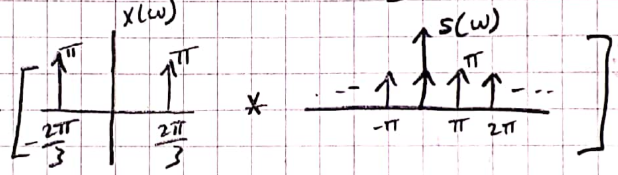
a) x(n)

b) X(w) çiz

c) y(t)

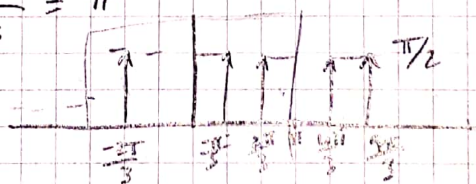
a) $t \to n.T_s = 2n$ $\quad x(n) = \cos \frac{4\pi}{3} n$

b) $X_s(\omega) = \frac{1}{2\pi}$



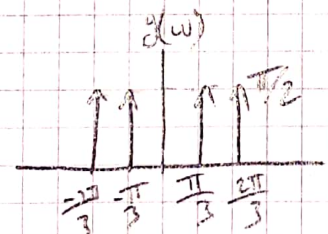
$\omega(s) = \frac{2\pi}{T_s} = \pi$

X(s) =



$= \frac{1}{\cancel{2\pi}} \cdot \cancel{\pi} \cdot \pi = \frac{\pi}{2}$

c)

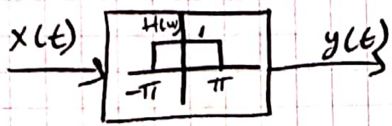


$y(t) = \frac{1}{2}\cos\frac{\pi}{3} + \frac{1}{2}\cos\frac{2\pi}{3}$

Yar oldu - final sorusu

Örnek! $X(\omega) = \pi\delta\left(\omega - \frac{3\pi}{2}\right) + \pi\delta\left(\omega + \frac{\pi}{2}\right)$

$T_s = 1 \ Hz$

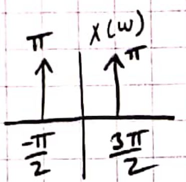


a) $X_s(\omega)$ çiz

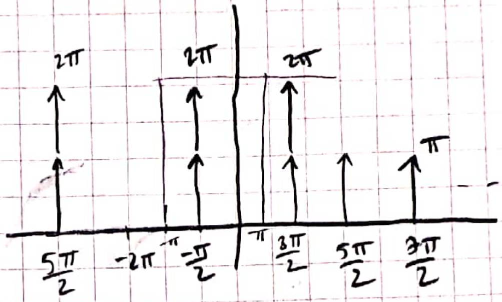
b) $y(t)$ -?

c) $x(n) = ?$

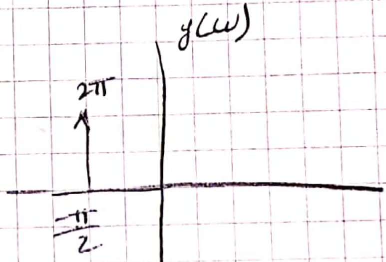
$X_s(\omega) = \frac{1}{2\pi}\Big[$



$*$



$Xs(\omega) = \frac{1}{2\cancel{\pi}} \cdot \pi \cdot \cancel{2}\pi = \pi$



$t \Rightarrow n.T_s = n$

$x(n) = \frac{1}{2}\delta\left(n - \frac{3\pi}{2}\right) + \frac{1}{2}\delta\left(n + \frac{\pi}{2}\right)$

$y(t) = e^{-j\frac{\pi}{2}t}$

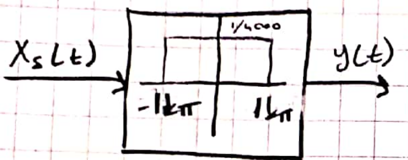
$y(\omega) = \delta\left(t + \frac{\pi}{2}\right)$

End of 18.12.2019

25.12.2019

$N = \frac{2\pi k}{w_0 T_s}$ → sonucu tamsayı ise, periyodiktir.

Örnek: $x_a(t) = A\cos(6000\pi t)$ işareti 4ms periyotla örneklenecektir.

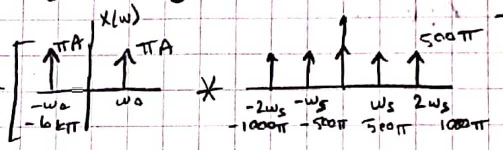


a) $X_s(t)$'nın frekans spektrumunu çiziniz.

b) $y(t) = ?$

c) $x(n)$ yada $y(n) = ?$

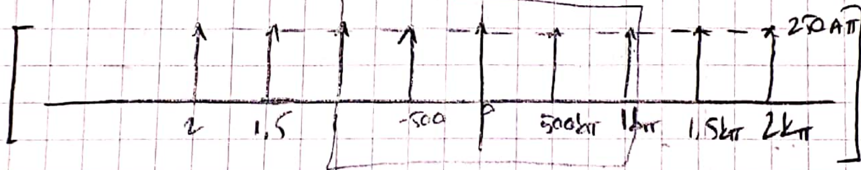
a) $X_s(w) = \frac{1}{2\pi}\left[X(w) * S(w)\right] = \frac{1}{2\pi}\Big[$



$\Big]$

$T_s = 4ms \Rightarrow w_s = \frac{2\pi}{T_s} = \frac{2\pi}{4 \cdot 10^{-3}} = 500\pi$

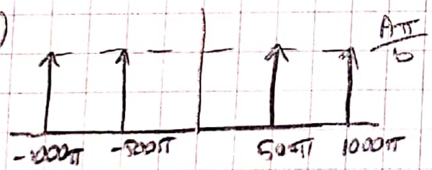
$= \frac{1}{2\pi}\Big[$



$\Big]$

$\frac{1}{2\pi} \cdot \pi \cdot A \cdot 500\pi = 250\,A\pi$

b)



$250\,A\pi \cdot \frac{1}{6000} =$

$y(t) = \frac{A}{32} + \frac{A}{16}\cos 500\pi t + \frac{A}{16}\cos 1000\pi t$

c) $t - n.T_s = 4.10^{-3}.n$

$$\boxed{x(n) = A.\cos(24\pi n)}$$

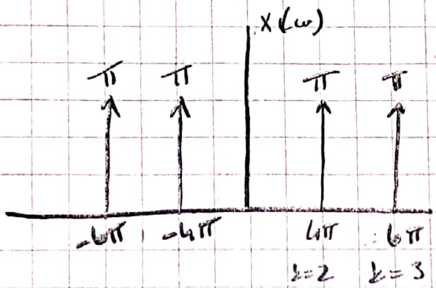
örnek! $X(\omega)=\pi[\delta(\omega-6\pi)+\delta(\omega-4\pi)+\delta(\omega+4\pi)+\delta(\omega+6\pi)]$

a) $w_0=?$

b) katsayılar=?

c) $x(t)=?$

a) EBOB $(6\pi, 4\pi) = 2\pi$



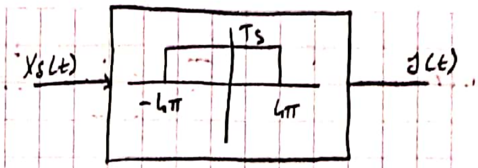
$2\pi . a_k = \pi$

$a_k = \frac{1}{2}$

b) $a_2 = a_{-2} = a_3 = a_{-3} = \frac{1}{2}$

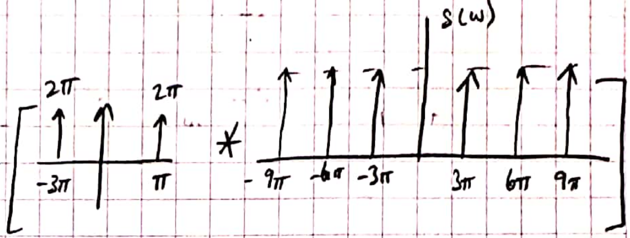
c) = $x(t) = \cos 4\pi t + \cos 6\pi t$

Örnek: $x(t) = e^{j\pi t} + e^{-j3\pi t}$



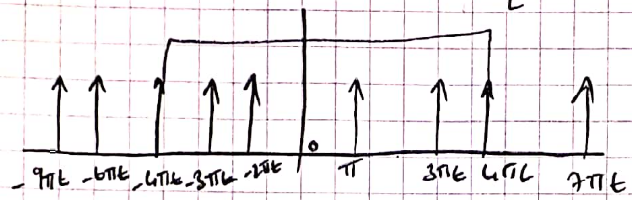
$T_s = \frac{2}{3}$ sn $\qquad \omega_s = \frac{2\pi}{T_s} = 3\pi$

$X_s(\omega) = \frac{1}{2\pi}\left[X(\omega) * S(\omega)\right] = \frac{1}{2\pi}\Big[$

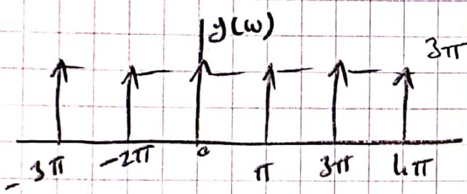


$\Big]$

$=$



$\frac{1}{2\pi} \cdot 2\pi \cdot 3\pi = 3\pi$



$3\pi \cdot \frac{2}{3} = 2\pi$

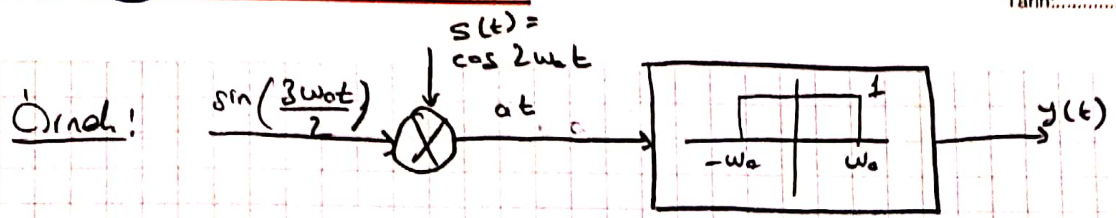
$$Y(\omega) = 2\pi\left[\delta(\omega) + \delta(\omega - 3\pi) + \delta(\omega + 3\pi) + \delta(\omega - 4\pi) + \delta(\omega + 2\pi)\right]$$

$$y(t) = \frac{1}{2\pi}\int Y(\omega)\, e^{j\omega t}\, d\omega$$

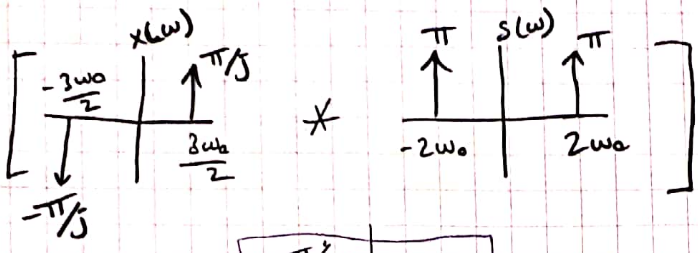
$$= \left[e^{j0t} + e^{j3\pi t} + e^{-j3\pi t} + e^{j4\pi t} + e^{-j2\pi t}\right]$$

$$= 1 + 2\cos 3\pi t + e^{j4\pi t} + e^{-j2\pi t}$$

Örnek!

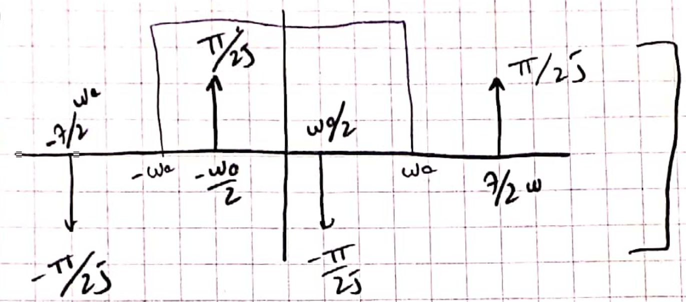


$a(\omega) = \frac{1}{2\pi}$ [



]

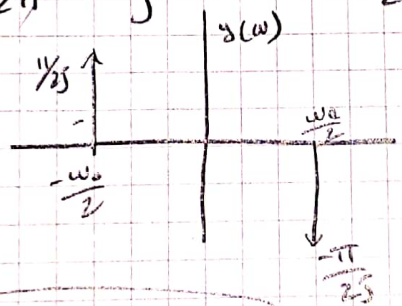
$= \frac{1}{2\pi}$ [



]

$$= \frac{1}{2\pi} \cdot \frac{\pi}{j} \cdot \pi = \frac{\pi}{2j}$$

$y(\omega) =$



$$\boxed{y(t) = -\frac{1}{2} \sin \frac{w_0}{2} t}$$

2. yol

$$y(\omega) = -\frac{\pi}{2j}\,\delta\left(\omega - \frac{w_0}{2}\right) + \frac{\pi}{2j} \cdot \delta\left(\omega + \frac{w_0}{2}\right)$$

$$y(t) = \frac{1}{2\pi} \int y(\omega)\, e^{j\omega t}\, d\omega$$